

restricted to 4m² in accordance with race regulations. Therefore they use 260 cells of 125mm × 125mm dimensions. These cells are arranged serially into two strings that are connected in parallel to a set of maximum power point trackers (MPPTs).

Motor. ARKA is driven by Mitsubishi M-2096 D III brushless DC in-hub motor; 2096 signifies that it consumes 2.0kW and works best at 96V. The motor is the main consumer of the power generated by solar panels. If the power required by the motor is more than what the solar array can produce at a moment, the battery works as a backup and supplies the remaining required power to it. As the solar car has a single-wheel drive system, this motor generates enough power to drive the car beyond 90kmph by only taking 2kW power, which is equivalent to the power needed for a hair dryer.

Telemetry system. ARKA is a connected car—a vehicle that can optimise its own operation and maintenance as well as the convenience and comfort of passengers using onboard sensors and Internet connectivity.

The team is completely on board with the idea of a connected car and believes in its potential as our future. ARKA embodies this philosophy, being a race car made for the most gruelling and competitive events in

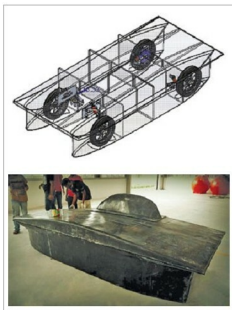


Fig. 5: Car body design

the solar car racing circuit.

Data acquisition occurs from the system with respect to various electrical and mechanical parameters. The state of charge of the battery is measured from the BMS. The vehicle's speed (rpm) is measured from a pulse-width signal from the motor controller, and the distance travelled is duly calculated. Temperature of the battery pack, BMS and photovoltaic solar cells is also measured. Incident solar irradiance is measured with the help of an external pyranometer, which will eventually be implemented onboard.

Data that is sensed in real time cannot be logged onboard due to constraints in the form of compute, memory and speed. This compels the team to log the data and store it in the cloud where it can be accessed

by anyone, anywhere and processed directly online. The heart of the sensing and data acquisition is BeagleBone Black, which also functions as the processing unit for the dashboard display.

All data is sent to a chase vehicle, which follows the prototype, and is then uploaded onto the cloud. IBM helps with data analytics and storage. Team RVCE uses IBM's proprietary IBM Cloud (Bluemix) platform. Data is stored in Cloudant NoSQL databases to handle the large amounts of data that are constantly uploaded. Analysis of the received data is done using another IBM service (data science experience), which uses Apache Spark and Python to parse through the data and implement machine learning techniques to come up with an optimal speed at which the car should be driven to maximise battery efficiency and minimise race time.

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To retain customers, Feedify's feedback tools ensure that the subscribers

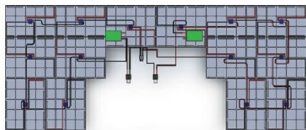


Fig. 3: Solar array maximum power point trackers

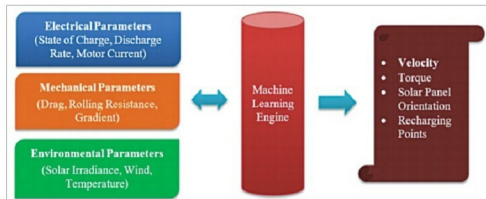


Fig. 4: Description of primary telemetry tasks